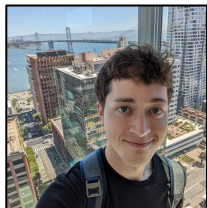


Culture Cartography: Mapping the Landscape of Cultural Knowledge

czyems@stanford.edu

Caleb Ziems



William Held



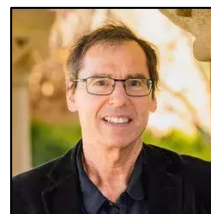
Jane Yu



Amir Goldberg



David Grusky



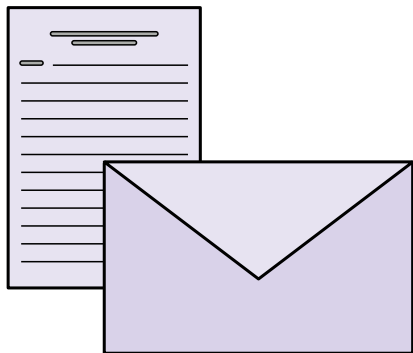
Diyi Yang



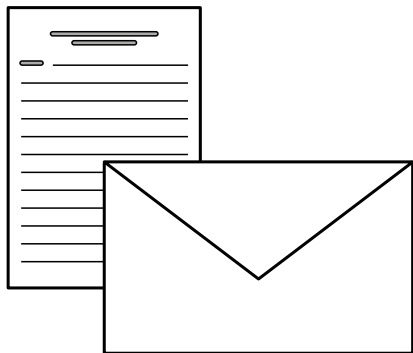
SALT

LLMs need **culture-specific** knowledge

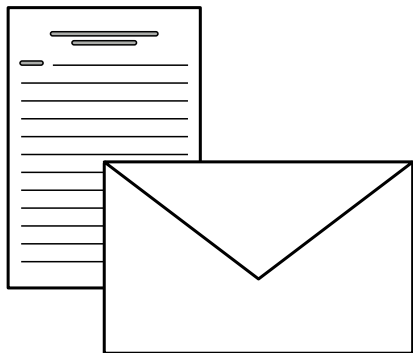
LLMs need **culture-specific** knowledge



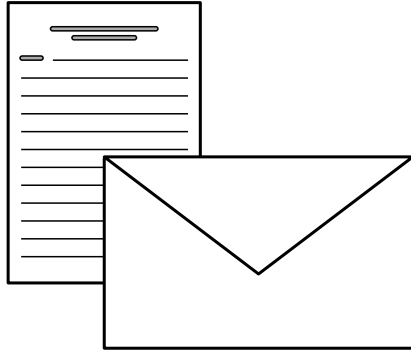
LLMs need **culture-specific** knowledge



LLMs need **culture-specific** knowledge



LLMs need **culture-specific** knowledge



Kuah Beulangong

soup

Aceh, Indonesia



starfruit

star
anise



what ingredients should I add?



banana
stems

jackfruit



Misalignment



SALT



Knowledge Gaps

1. **OOD** for current models
2. **Representative** of users' needs

Methods for Eliciting Cultural Knowledge:

Traditional Annotation



May Not Represent Users



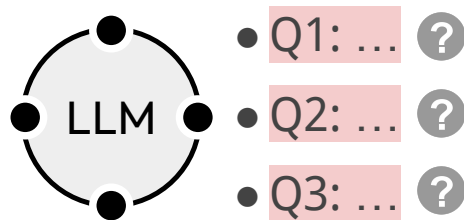
Methods for Eliciting Cultural Knowledge:

Traditional Annotation

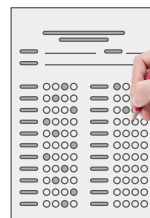
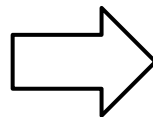
(*Single Initiative*)



May Not Represent Users



Researcher Produces
Challenging Questions



- A1
- A2
- A3

Humans **Passively**
Answer Questions

Methods for Eliciting Cultural Knowledge:

Knowledge Extraction

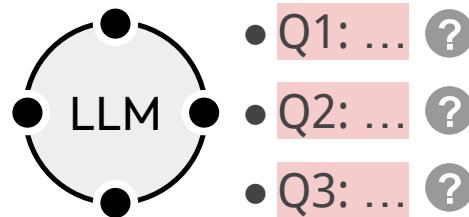
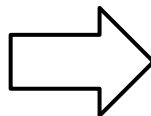
(*Single Initiative*)



May Not Challenge LLMs



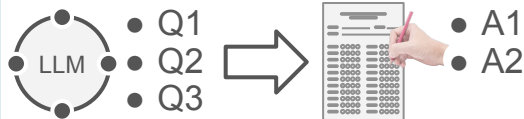
Humans Actively
Produce Data



LLM Formats a
Benchmark

May Not Represent Users

Traditional
Annotation



Methods for Eliciting Cultural Knowledge:

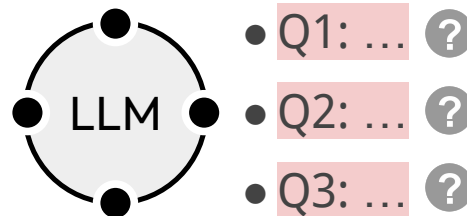
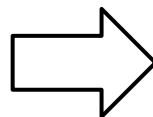
Knowledge Extraction

(*Single Initiative*)

! May Not Challenge LLMs !
! Prone to Test-Set Contamination !



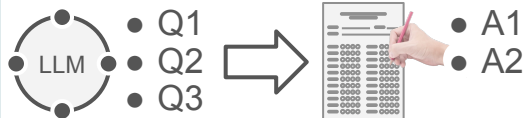
Humans Actively
Produce Data



LLM Formats a
Benchmark

May Not Represent Users

Traditional
Annotation



Culture Cartography

(*Mixed Initiative*)

elicit **data** that is...

1. **Representative** of users' needs....

the **human**
steers and edits

2. **OOD** for current models

the **LLM**
identifies
challenging
questions

May Not Represent Users



May Not Challenge LLMs





Culture Cartography

A

Seed Topic:

model generated; human editable

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events



wedding
practices





Culture Cartography

A Seed

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events



wedding
practices



B Question:

can be edited, regenerated, or deleted

List any customs or traditions
related to the presentation of
Christmas gifts in
Indonesian culture

List any adaptations or changes in
Indonesian gifting customs that have
occurred over time due to social or
technological advancements.



Culture Cartography

A Seed

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events

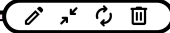


wedding
practices

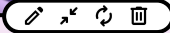


B Question:

can be edited



List any customs or traditions
related to the presentation of
~~Christmas~~ Idul Fitri gifts in
~~Indonesian~~ Javanese culture



List any adaptations or changes in
Indonesian gifting customs that have
occurred over time due to social or
technological advancements.



Culture Cartography

A Seed

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events

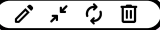


wedding
practices

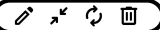


B Question:

can be  **edited**



List any customs or traditions related to the presentation of Idul Fitri gifts in Javanese culture



List any adaptations or changes in ~~Indonesian~~ **Javanese** gifting customs that have occurred over time due to social or technological advancements.





Culture Cartography

A Seed

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events



wedding
practices



B Question

List any customs or traditions related to the presentation of Idul Fitri gifts in Javanese culture

List any adaptations or changes in Javanese gifting customs that have occurred over time due to social or technological advancements.

C Answer:

uncertain answers are highlighted

The use of e-wallets and QR Codes has become increasingly popular in Indonesia's urban areas.

During weddings the traditional "hantaran" (gift parcels) are now often curated by professional services.

For Eid al-Fitr, giving money or "duit lebaran" in decorative envelopes has remained a tradition, but now the designs are influenced by popular online trends.



Culture Cartography

A Seed

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events



wedding
practices



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Culture Cartography

A Seed

artistic
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family
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gift
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national
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wedding
practices



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During weddings the traditional "hantaran" (gift parcels) are now often curated by professional services.

For Eid al-Fitr, giving money or "duit lebaran" in decorative envelopes has remained a tradition, but now the designs are influenced by popular online trends.

D Follow-Up Q/A

List the traditional methods of preparing and presenting "hantaran" before the rise of professional services.

List the reasons why families might choose professional services to curate their "hantaran" for weddings

List the occasions, other than weddings, where "hantaran" might be a traditional practice in Indonesian culture.

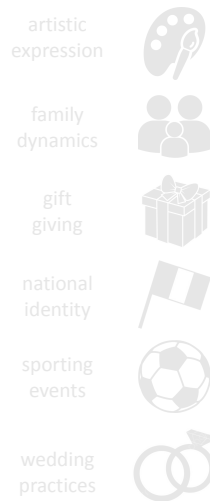
Time Efficiency
Professional process, saving significant time.

Expert Presentation
Professional curators create stunning and authentic displays.

Quality Assurance
Items selected by professionals ensure high quality, even for family members.

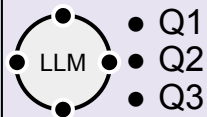


A Seed



Hypothesis 1: Culture Cartography data better challenges models than:

Traditional Annotation



• Q1
• Q2
• Q3



• A1
• A2

Corollary 1: LLMs recall less of the  Culture Cartography data than they recall Traditional Annotations.

Follow-Up Q/A



traditional methods
paring and presenting
"hantaran" before the rise
professional services.



reasons why families
choose professional
to curate their
"hantaran" for weddings



occasions, other
than weddings, where
"hantaran" might be a
traditional practice in
Indonesian culture.

Time Efficient
Professional
process, saving
significant

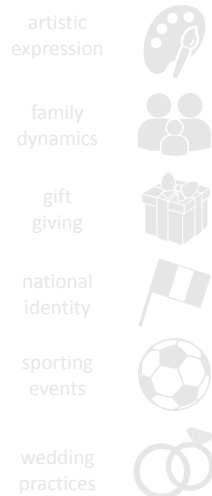
Expert Professional
stunning and
reflect modern
authenticity

Quality Assurance
Items selected by
professionals ensure
quality, especially for
family



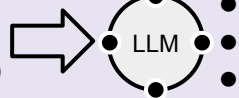
Culture Cartography

A Seed



Hypothesis 2: unlike:

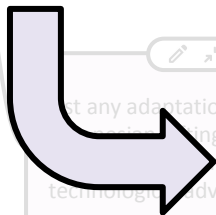
Knowledge
Extraction



Q1: ... A1
Q2: ... A2
Q3: ... A3



Culture Cartography data is *Google-Proof*



Corollary 2: search-enabled LLMs will not outperform search-disabled LLMs on  **Culture Cartography** data.

Follow-Up Q/A

traditional methods
paring and presenting
"man" before the rise
essional services.

During weddings the traditional
List the reasons why families
choose professional
s to curate their
"man" for weddings

occasions, other
eddings, where
"man" might be a
nal practice in
ndonesian culture.

Time Efficient
Profession
process, sa
significant

Expert Pre
Profession
stunning a
reflect mo
authentic

Quality As
Items sele
profession
quality, en
famil



CultureExplorer

A Seed

artistic
expression



family
dynamics



gift
giving



national
identity



sporting
events

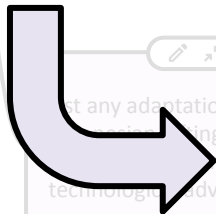


wedding
practices



Hypothesis 3: Culture Cartography

data is aligned with existing operationalizations of culture in prior benchmarking efforts.



Corollary 3: fine-tuning on  Culture Cartography data will boost model performance on prior benchmarks.

Follow-Up Q/A

traditional methods
paring and presenting
"ran" before the rise
essional services.

During weddings the traditional
List the reasons why families
choose professional
s to curate their
"ran" for weddings

occasions, other
weddings, where
"ran" might be a
onal practice in
ndonesian culture.

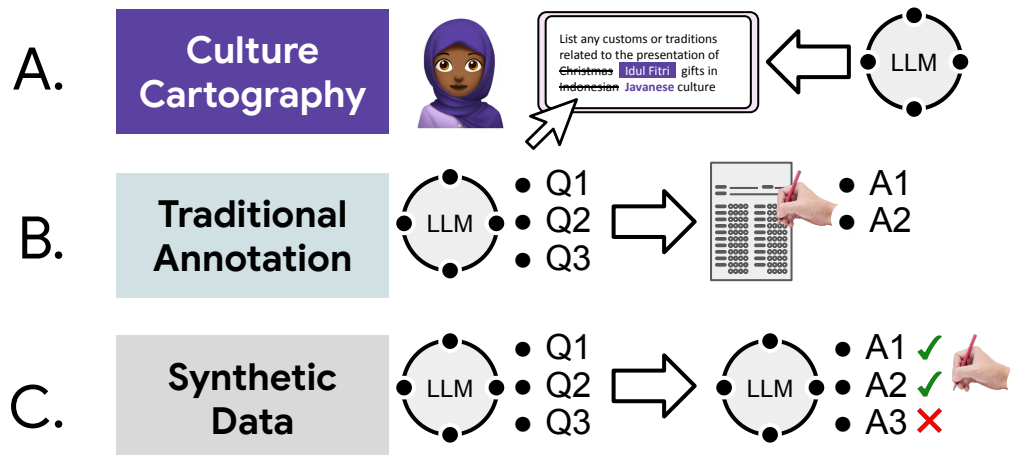
Time Efficient
Profession
process, sa
significant

Expert Pre
Profession
stunning a
reflect mo
authentici

Quality As
Items sele
profession
quality, en
famil

Testing Hypotheses

Data Collection: (via **upwork**)



Testing Hypothesis 1

Culture Cartography is better than Traditional Annotation

A.

Culture
Cartography

B.

Traditional
Annotation

C.

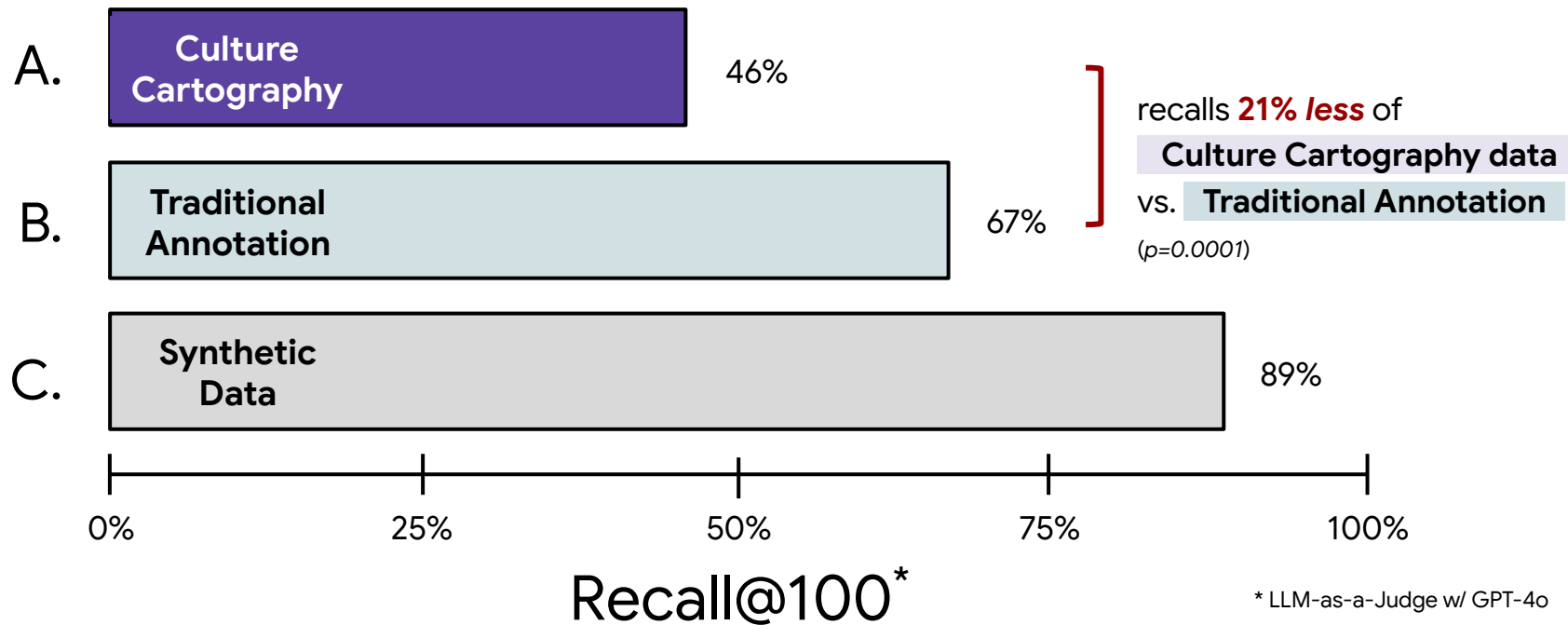
Synthetic
Data

$$\text{Recall@100}^* = \frac{|\{\text{gold answers}\} \cap \{\text{model answers}\}|}{|\{\text{gold answers}\}|}$$

Testing Hypothesis 1

Culture Cartography is better than Traditional Annotation

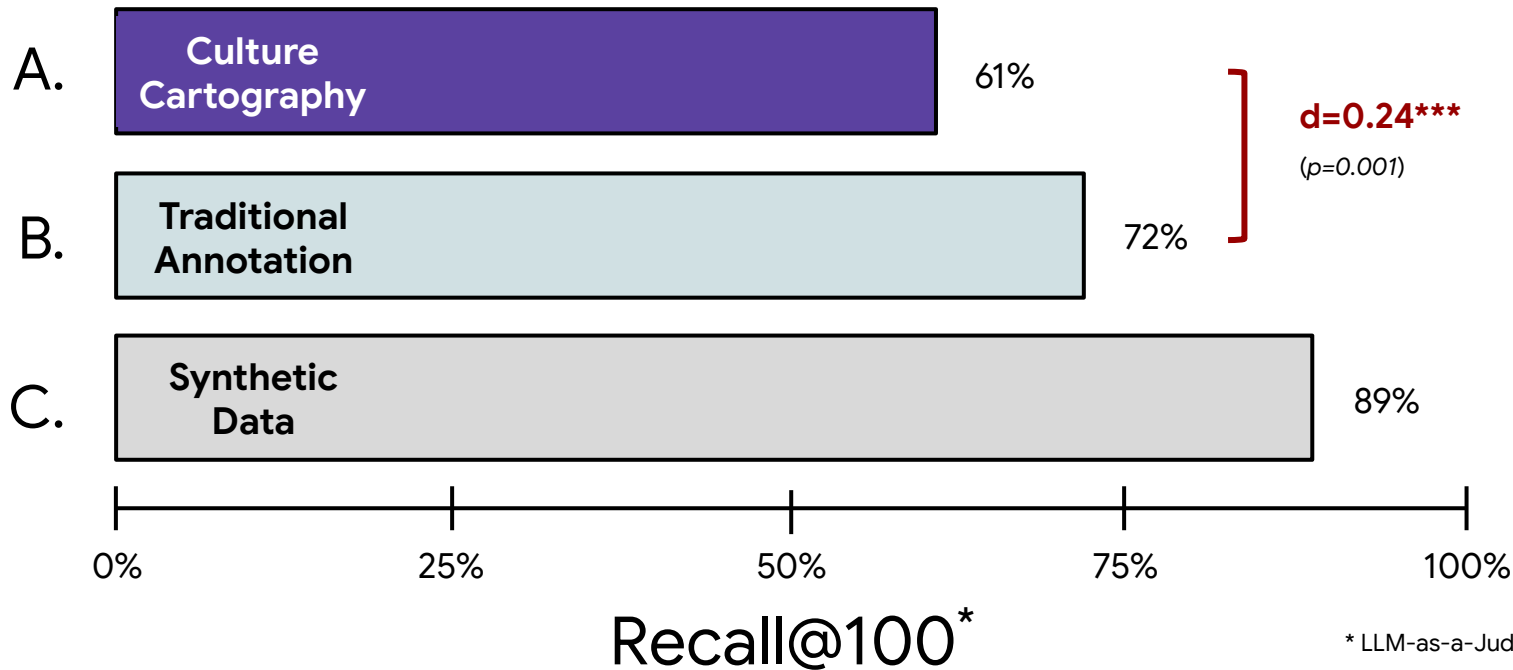
Results for:  Qwen-2-72B



Testing Hypothesis 1

Culture Cartography is better than Traditional Annotation

Results for:  Llama-4 Maverick

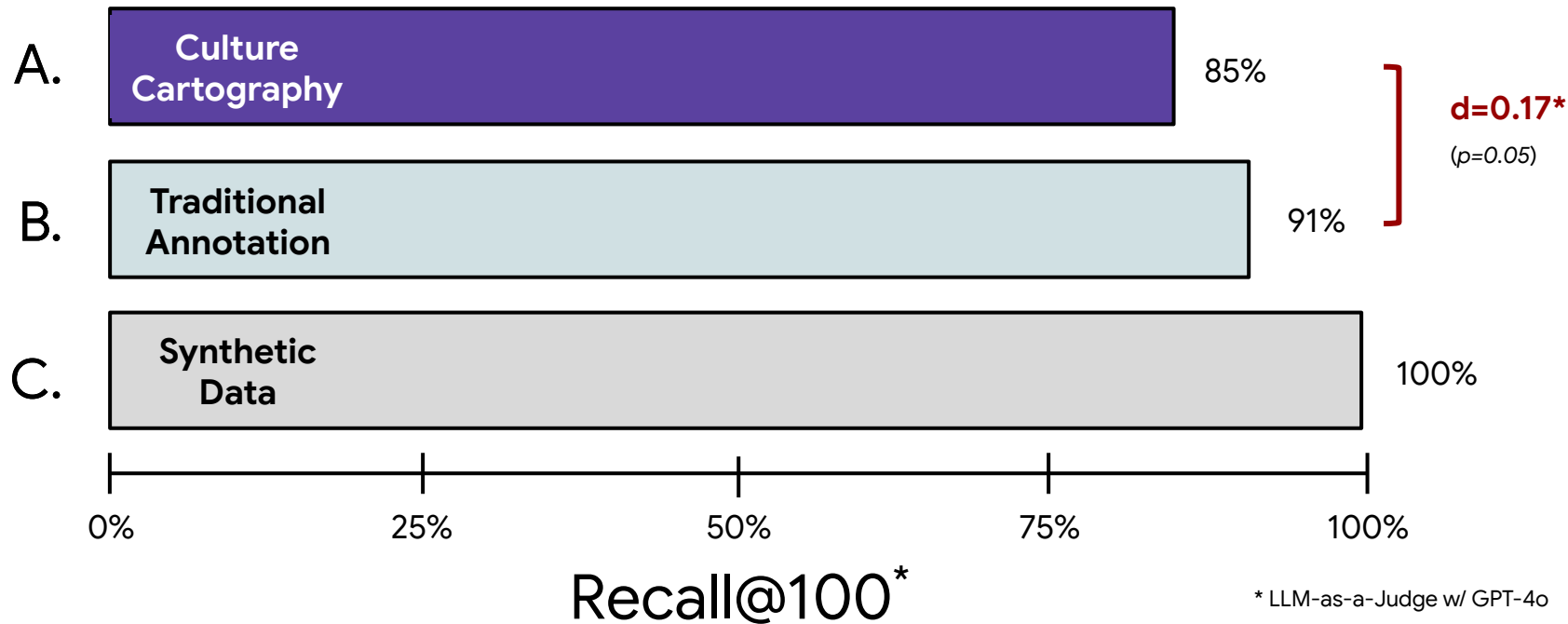


* LLM-as-a-Judge w/ GPT-4o

Testing Hypothesis 1

Culture Cartography is better than Traditional Annotation

Results for:  DeepSeek R1

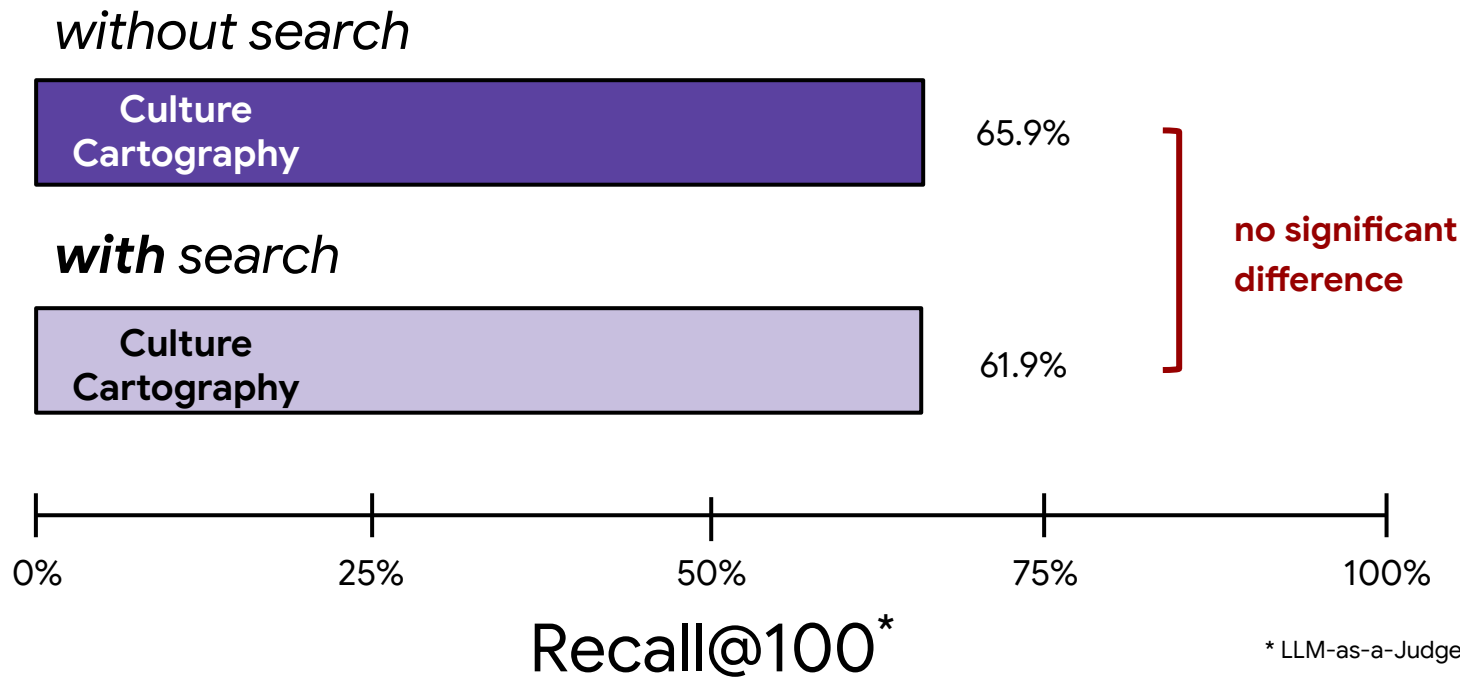


* LLM-as-a-Judge w/ GPT-4o

Testing Hypothesis 2

Culture Cartography is better than Knowledge Extraction

Results for:  GPT-4o



* LLM-as-a-Judge w/ GPT-4o

Testing Hypothesis 3

Culture Cartography is aligned with the objectives of the field

BLEnD

Traditional Annotation

What do people eat on their **birthday** in **South Korea**?

미역국
(Seaweed
Soup)

케이크
(Cake)

갈비
(Galbi)

What is the most common spice/herb used in dishes from **Greece**?

- A. Black Pepper
- B. Cumin
- C. Epazote
- D. Oregano



CulturalBench

Red Teaming

Human review after AI generation on question

What do Singaporeans usually use to reserve seats?

- a. Tissue packet
- b. Wallet
- c. Sunglasses
- d. Book

Ans: A
Confidence : 56%



Human edit the question until successfully trick the chatbot

Testing Hypothesis 3

Culture Cartography is aligned with the objectives of the field

Llama-3.1-8B 

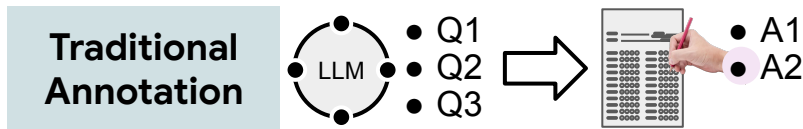
BLEnD

CulturalBench

					
Vanilla Model		56.0	66.5	50.0	46.2
SFT+DPO	Traditional Annotation	59.3	73.2 ***	50.0	65.4 *
	Culture Cartography	62.5 ***	73.6 ***	63.6 *	65.4 *

Conclusions

 **Culture Cartography** data is more **challenging** than



 **Culture Cartography** data is **Google-proof**, unlike



 **Culture Cartography** is **aligned** with the objectives of the field

BLEnD

CulturalBench

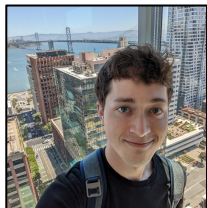
Traditional Annotation

Red Teaming

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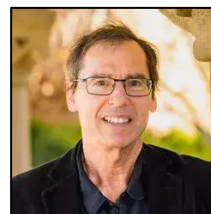
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SALT